

**SECOND YEAR SOFTWARE ENGINEERING
SPRING SEMESTER ASSIGNMENT 1, 2026**

CLO 1

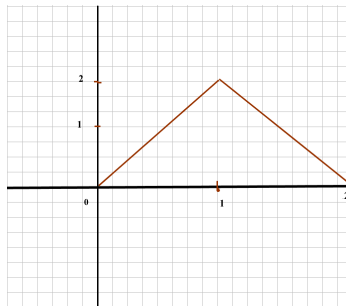
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Question 1.....

Let a robotic arm's angular position $y(t)$ satisfies $y'' + 3y' + 2y(t) = 10e^{-2t}$. If the arm starts from rest at position $y(0) = 0.5$ and $y'(0) = 0$, determine $y(t)$ using the Laplace Transform method.

Question 2.....

Write the following function in terms of unit step function and then find the Laplace transform.



Question 3.....

Find the Laplace transform of $f(t) = \begin{cases} 2t + 1, & 0 \leq t < 2, \\ 3t, & t \geq 2, \end{cases}$

Question 4.....

Use the standard formula to derive the Laplace transform of $f(t) = \cos(t + 2\pi)$.

Question 5.....

Find

(i) $\mathcal{L}^{-1} \left\{ \frac{e^{-2s}}{s^2 + 4s + 8} \right\}$

(ii) $\mathcal{L}^{-1} \left(\frac{e^{-s}}{s(s^2+1)} \right)$

Question 6.....

Solve following differential equations

(i) $y'' - 10y' + 9y = 5t$, $y(0) = -1$ $y'(0) = 2$

(ii) $2y'' + 3y' - 2y = te^{-2t}$, $y(0) = 0$ $y'(0) = -2$